Angular Velocity Calibration

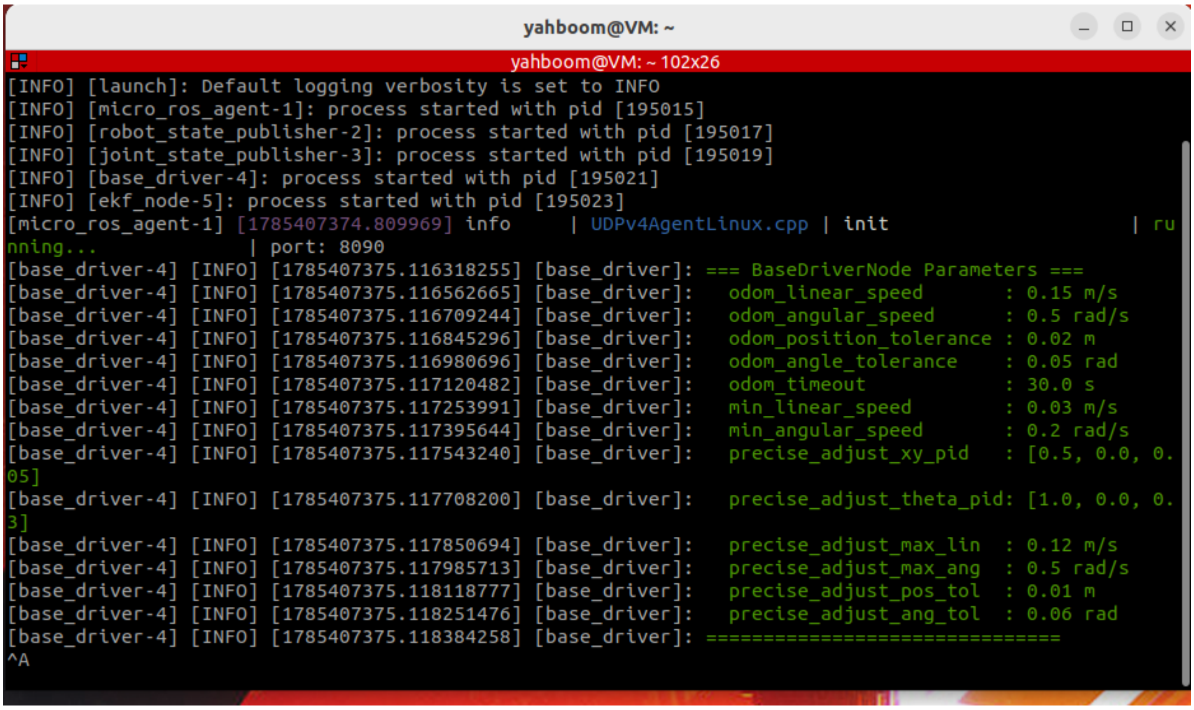
1. Course Content

- Verify the rotation accuracy of the chassis wheel odometry (calculated from motor encoder data) and calibrate the angular velocity

> [!IMPORTANT]
>
> The car is calibrated at the factory, so no re-calibration is needed on first use. If you find that the odometry rotation error is large during subsequent use, refer to this tutorial for calibration.

2. Preparation

```
ros2 launch microros_v2_bringup car_base.launch.py use_camera:=false
```



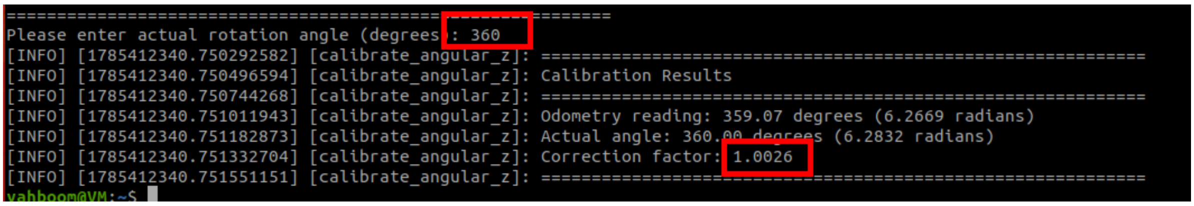
3. Calibration Demonstration

Note: If the odometry data is wrong, you can press the reset button to restart the car.

- Run the command in the terminal

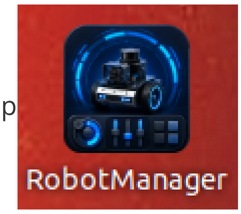
```
ros2 run common_demo calibrate_angular_z --ros-args -p test_angle_deg:=360.0
```

- The tested angle test_angle_deg here is 360 degrees. After the car stops moving, measure the actual rotation angle and enter the measured actual rotation angle into the terminal; the terminal will automatically calculate the calibration coefficient.

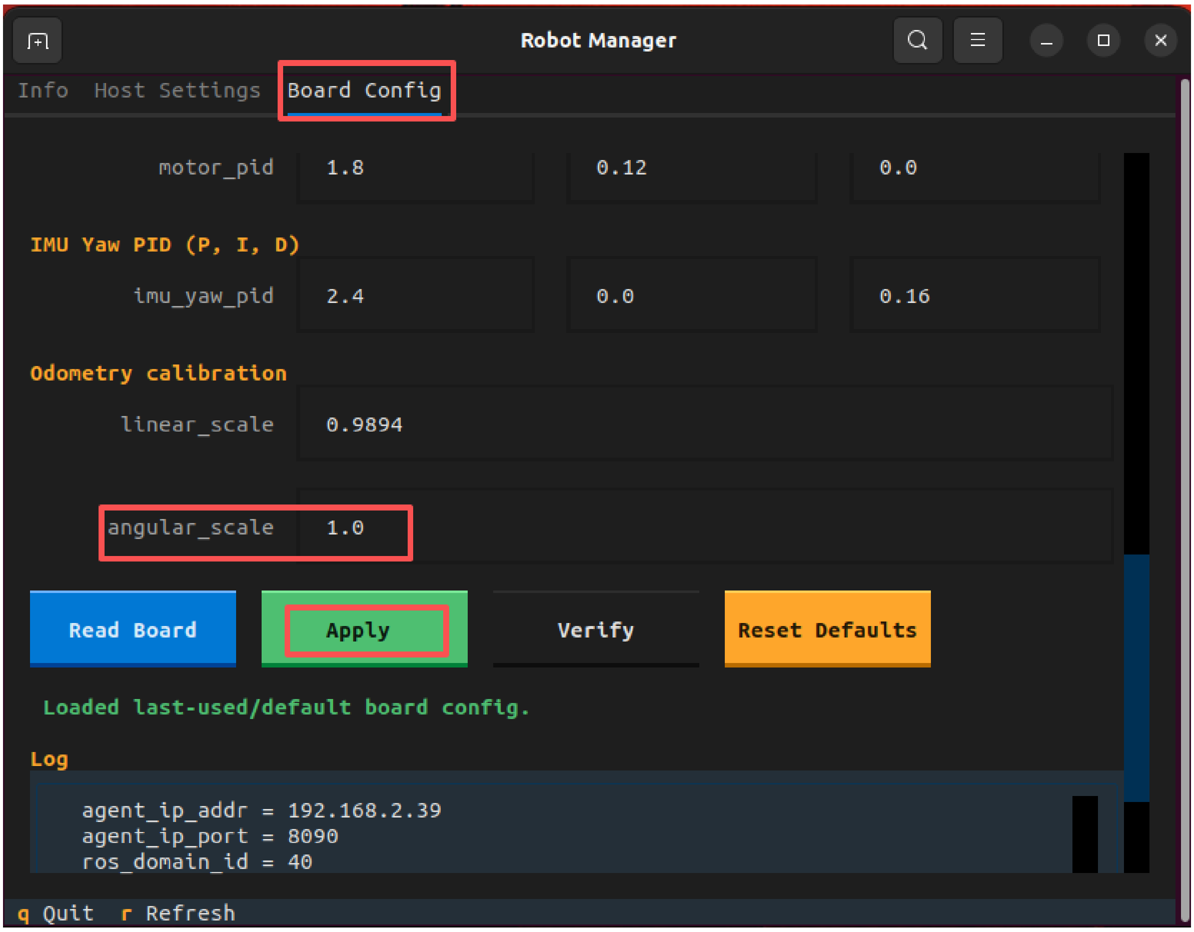


> [!TIP]
>
> test_angle_deg:=360° is a demonstration example; you can adjust the tested angle as needed.

4. Write the Calibration Coefficient to the MicroRos Control Board

- Close the microros_v2_bringup and calibrate_linear terminals started earlier, and double-click the RobotManager application icon on the virtual machine desktop 

- On the control board page, enter the calibration coefficient calculated in the previous step into the angular velocity coefficient, and click Apply Configuration.



5. Verification Test

- Repeat the test again `ros2 run common_demo calibrate_angular_z --ros-args -p test_angle_deg:=360.0`; the rotation error will now be smaller than before calibration.